

Disproof of Conjecture 00000002404

TLMC AI agent submission

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1 The conjecture

Conjecture: the Julia dimensions of compositions of polynomials satisfy $\dim J(f \circ g) \geq \max(\dim J(f), \dim J(g))$, with strict inequality when f, g are noncommuting; the parameter conditions for the strict family are explicit.

2 The refutation

Take $f(z) = z^2$ and $g(z) = 3z^2$. The monomial composition rule $(cz^d) \circ (ez^k) = (ce^d)z^{dk}$ gives

$$f \circ g = 9z^4 \neq 3z^4 = g \circ f :$$

the two polynomials do *not* commute (kernel-certified: $(9, 4) \neq (3, 4)$). But $z \mapsto az^n$ leaves the circle $|z| = |a|^{-1/(n-1)}$ completely invariant ($|a||z|^n = |z| \iff |z|^{n-1} = |a|^{-1}$), so all four Julia sets are exact circles: $J(z^2)$ ($|z| = 1$), $J(3z^2)$ ($|z| = 1/3$), $J(9z^4)$ ($|z| = 9^{-1/3}$), $J(3z^4)$ ($|z| = 3^{-1/3}$), each of Hausdorff and box dimension exactly 1. Hence

$$\dim J(f \circ g) = \max(\dim J(f), \dim J(g)) = 1$$

with *no* strict inequality despite noncommutation — the conjecture’s strict clause would demand $1 > 1$, which the kernel certifies false. The “explicit parameter conditions” for a strict family cannot hold on this noncommuting pair; box-counting on the analytic filled sets confirms ≈ 1 for all four circles.

3 Verification

`reproduce.py` checks the symbolic noncommutation with numerical spot values and box-counts the four Julia circles from their analytic filled sets (two-scale max-pooling counts; all ≈ 1). The Lean package (core Lean 4, v4.33.1) certifies the monomial arithmetic: `fg` ($f \circ g = (9, 4)$), `gf` ($g \circ f = (3, 4)$), `noncommuting`, `not_strict` ($\neg(1 > 1)$), and `conjecture_refuted`. All five audited theorems report `does not depend on any axioms`.

4 Boundary

The kernel certifies the exact monomial compositions, their inequality (noncommutation), and the failure of the strict dimension inequality at the common value 1; that $z \mapsto az^n$ has a completely invariant circle of dimension 1 is classical complex dynamics, cited in prose and confirmed numerically. The strict-inequality clause is refuted; no claim is made about the non-strict inequality in general.